

MAJOR CONTRIBUTORS TO THIS REPORT



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car (Mahindra e2o); now this is a feat achieved," he explains delightfully.

Apart from ongoing research on battery chemistries and battery management systems, Mahindra Reva is also working with CECRI, Karaikudi (a CSIR lab) to research on lithium-ion batteries that will help maximise battery life, optimise battery systems design, predict failures, etc.

Gupta, however, feels that despite EVs emerging as a major trend in the smart vehicles space, these are yet to be widely accepted due to dependence on infrastructure and policies around the same.

"Infrastructure has always been a chicken and egg problem for us. But, as we progressed along the learning curve, we have been able to solve the complicated equation by coming up with our own charging points at strategic locations. We are now at a point that we are working on a network of fast-charging stations so you can charge over a cup of coffee and move on," answers Maini. He also explains that EVs are for people with fixed and predictable travel patterns, so they can comfortably charge at home or office and sail through their needs.

"To reduce the anxiety in the EV users, we are already partnering with malls and retailers to build EV charging ports into their parking lots, but government support is important to build a critical mass of users and suppliers and smooth out current regulatory roadblocks in this space," he adds. For instance, we already have over 300 charge ports all over the country and are planning to extend it to other cities. We are hopeful that the new

National Mission for Electric Mobility will help drive this. We can envision a not-so-distant future (it already exists in other parts of the world) where you can take your EV to a fast-charging station, get an 80 per cent charge in 15 minutes, and be off to your next destination."

More updates.

► A start-stop system is one that can automatically shut down and restart the internal combustion engine to reduce engine idling time and thereby fuel consumption too. Whilst this was initially popular only in cars with a hybrid electric powertrain, now it is being adopted in non-electric cars too.

► Recently, Toyota launched its FCV hydrogen electric vehicle, expected to come to market next year. In this innovative car, hydrogen and oxygen combine to create electricity and water so that a zero emission target can be achieved.

► Ford and Georgia Tech have together developed the Ford C-MAX Energi Solar, a hybrid vehicle that can charge its battery using a solar panel fitted on its roof. The car includes autonomous features (such as that used for parking) to follow the sun's movements and adjust the panel direction throughout the day. Ford claimed at the demo that a day's worth of sunlight will provide the same amount of power and performance as the plug-in variant of C-MAX Energi, able to return a fuel efficiency of 2.4 litres per 100km and a range of almost 1000km.

► Honda, the first to put a fuel cell vehicle (FCX Clarity) in the hands of consumers, recently unveiled its new Honda FCEV Concept sketch, a fu-

turistic and aerodynamic design of its next-generation fuel-cell EV launching in 2015. This will be a step towards a zero-emission future. Honda's alternate energy range includes Accord Hybrid, Accord Plug-In Sedan, Honda Fit EV and the Civic Natural Gas.

► Another rather off-beat step towards environment-friendly technology is Honda Motor's agreement with TDK Corporation and Japan Metals & Chemicals to work towards the reuse of a rare earth metal extracted from nickel-metal hydride batteries in hybrid vehicles for magnets of new hybrid vehicle motors. This will reduce the environmental footprint of mobility.

My car, my future

Technologies abound, for easier, safer and more sustainable mobility. However, everything is not good. As scientists always believe, a technology is to be evaluated more by its impact on society than the technological merits. Cars are easier to drive, good news! But, I live next door to a school and am startled by the increase in the number of moms dropping individual kids in cars, because they are easier to drive and park than two-wheelers! Where are the days when most kids used to travel by bus or cycle to school? How are we going to manage fuel availability and pollution in the near future? By the time hybrid and electric cars hit a mass-market pricing and are widely adopted, will not greater harm be done than good?

Hands-free messaging, complete integration of wearable tech and smartphone apps with car dashboards and in-car media centres are all very snazzy, but a recent *Scientific American* article shows that hands-free texting is still not safe as it distracts the driver. Will driver assistance features be powerful enough to avoid accidents caused due to distraction?

Tech is a good thing, but for it to be sustainable, users need to exercise their prudence too, than being carried away by all that seems good! ●

The author is a technically-qualified freelance writer, editor and hands-on mom based in Chennai